

Luan Dinh

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SUMMARY

Computer Science (AI) student and AI engineer who ships rigorously engineered systems. Building a self-growing, brain-inspired AI engine on spiking neural networks with green multi-OS CI and 350+ releases, local LLM infrastructure, and the agent-orchestration workflow behind them, including custom Claude Code skills and hooks that keep AI agents coordinated and verified. Focus: honest metrics, tested code, software that ships.

EDUCATION

DePaul University

Sep 2022 – Dec 2026 (expected)

B.S. Computer Science, Artificial Intelligence · GPA 3.4 · Chicago, IL

- Relevant coursework: Machine Learning, Deep Learning, Computational Neuroscience (NEU 256, in progress), Software Projects, Computer Systems, Algorithms.

VNU-HCM High School for the Gifted

Aug 2019 – Jun 2022

GPA 3.3 · SAT 1410 · Ho Chi Minh City, Vietnam

PROJECTS

Spiking Neural Data Lake (SNDL)

Python

github.com/PVLuanDinh/Spiking-Neural-Data-Lake

- Neurally-inspired AI engine built on spiking neural networks that grows its own neurons: when it cannot solve a problem it adds neurons, passes them through a separate “liquid” reservoir network so they learn how they relate to what it already knows, then runs a sleep cycle that prunes and saves the result to one permanent data store it can recall from.
- System design: two cooperating networks inside a single durable data lake, trained with local, biologically plausible learning rules (STDP, e-prop) instead of backpropagation; a safety layer with an emergency stop on every robot-arm (SO-101) command; and a live dashboard for monitoring and demos.
- Research method: every engine component is tested as a pre-registered experiment with controls and statistical tests before it merges. Built in parallel by a team of AI agents I direct, with peer review and one integrator; the final milestone removes each connection to prove it is needed.
- ~390 automated self-checks plus pytest; GitHub Actions CI on Ubuntu and Windows with free-threaded Python 3.14; 350+ tagged releases.

Claude Odysseus Optimizer

Python

github.com/PVLuanDinh/claude-odysseus-optimization

- Builds a queryable knowledge graph across codebases via AST parsing and static analysis, returning subgraphs sized for LLM context windows.

agent-board (Claude Code skill)

Python, open source (MIT)

github.com/PVLuanDinh/agent-board

- Claude Code skill that coordinates multiple AI agents: shared message board, blind peer review, and a scorer that re-runs verification instead of trusting agent reports; plus custom hooks that improve AI workflows, installable on any Claude account in one command.

Claude Code Router

Local LLM infrastructure

- Set up and run the open-source claude-code-router, self-hosted, so AI-coding-client traffic goes to a local Ollama model instead of the cloud; installed globally as a CLI and used live for cost control and privacy.

Agentic Build-Loop Engineering

Methodology

- Repeatable research, build, verify (CI), ship loops, one safe slice per cycle: multi-agent orchestration routing subtasks by model tier, persistent cross-session memory, and knowledge-graph navigation. Nothing merges without passing the full CI gate.

EXPERIENCE

AI Engineer & Website Developer

Aug 2024 – Present

SAMOVIC Ltd. · Remote (Ho Chi Minh City, Vietnam)

- Built a customer-facing website with an AI assistant that answers FAQs and guides customers via hyperlinks, part of revitalizing an aging company's digital presence.
- Built an AI scanner that converts architectural sketches into rough, editable 3D models in Blender.
- Raised sketch-to-3D throughput from ~3 to 10+ models per day.

SKILLS

Languages: Python (Advanced), Java, JavaScript/TypeScript, C++, SQL, Assembly

AI / Machine Learning: Neurally-inspired AI, computational neuroscience (coursework), neural networks from scratch, deep learning, TensorFlow, PyTorch, snnTorch, spiking neural networks & neuromorphic computing, surrogate-gradient training, STDP, NLP & chatbots, GPU/CUDA training, fp16 mixed precision

LLMs & Agents: Local LLM serving (Ollama), LLM request routing, retrieval-augmented context, multi-agent orchestration, custom AI skills & hooks (Claude Code), AI build loops, prompt engineering

Software Engineering: System design & architecture, safety-critical control paths, stdlib-only tool design, GitHub Actions CI matrices, pytest, ruff, pre-commit, test-driven development, semantic versioning & releases, packaging (PyInstaller, Inno Setup), software product testing & QA

Web & APIs: React, TypeScript, HTML, CSS, REST APIs, OpenAI API integration

Tooling & Infrastructure: Git/GitHub, knowledge graphs & static analysis, VS Code, Blender scripting (sketch to 3D), Adobe Creative Cloud

Spoken Languages: English (Bilingual), Vietnamese (Native), French (Conversational)